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RETAIL SALES PERFORMANCE ANALYSIS

Data Analytics Internship Project Report

Project Overview

This project analyzes historical retail sales data to identify key performance indicators, sales trends, top-performing products, profitable and loss-making product areas, regional performance, customer segment performance, and growth patterns. The analysis combines Python-based data cleaning and exploratory data analysis with a Power BI dashboard for interactive visualization and business decision-making.

Problem Statement

A retail company wants to optimize its sales strategy but lacks sufficient insight into key performance indicators. The objective is to analyze sales data to identify trends, patterns, and areas for improvement, and to provide actionable recommendations for improving sales performance.

Objectives

- Analyze historical sales data.
- Identify key performance indicators such as best-selling products, peak sales periods, and customer segment performance.
- Identify sales and profitability trends using exploratory analysis.
- Create visualizations that communicate sales performance clearly.
- Provide actionable recommendations for improving sales performance.

Key Tasks

- Clean and preprocess the sales dataset.
- Check missing values, duplicate records, and data types.
- Perform exploratory data analysis (EDA).
- Analyze sales by product, category, sub-category, region, customer segment, and time.
- Analyze profitability and profit margin.
- Calculate year-over-year sales growth.
- Create visualizations and an interactive Power BI dashboard.
- Translate analytical findings into business recommendations.

Tools and Technologies

- Python
- Pandas and NumPy
- Jupyter Notebook
- Matplotlib
- Power BI Desktop
- DAX for Power BI measures

1. Data Preparation and Cleaning

The dataset was loaded into Python using Pandas. Data quality checks were performed before analysis.

- Missing values: 0
- Duplicate rows: 0
- Order.Date and Ship.Date were converted from object/text format to datetime format.
- Additional Month and Month_Name fields were derived from Order.Date for time-based analysis.
- The cleaned dataset was exported as Global_Superstore_Cleaned.csv for use in Power BI.

Key Performance Indicators

KPI	Result
Total Sales	12,642,905
Total Profit	1,467,457.29
Total Quantity Sold	178,312
Overall Profit Margin	11.61%

2. Exploratory Data Analysis and Findings

2.1 Product Performance

Apple Smart Phone, Full Size was the top-selling product by sales, generating 86,936 in sales. This identifies the product as an important revenue contributor.

2.2 Category Performance

Technology was the highest-performing category in both sales and profit. It generated 4,744,691 in sales and 663,778.73 in profit. This indicates that Technology is a major driver of the company's overall performance.

2.3 Regional Performance

Central was the strongest region in both sales and profit, generating 2,822,399 in sales and 311,403.98 in profit. The consistency between sales leadership and profit leadership makes Central a useful benchmark for understanding successful regional performance.

2.4 Customer Segment Performance

The Consumer segment generated the highest sales, with 6,508,141 in sales. This indicates that Consumer customers are the largest sales contributor among the available customer segments.

2.5 Monthly Sales Trend

December was the peak sales month, with 1,580,816 in sales, while February recorded the lowest monthly sales, with 543,768. This shows a substantial difference between peak and low sales periods and suggests an opportunity for seasonal planning.

2.6 Sub-Category Profitability

Copiers was the most profitable sub-category, generating 258,567.55 in profit. Tables was the least profitable sub-category, generating a loss of 64,083.39. This highlights the importance of evaluating profitability rather than relying only on sales volume.

2.7 Year-over-Year Sales Growth

The highest year-over-year sales growth was 27.2% in 2013. The lowest growth rate was 18.5% in 2012. Although both represent positive growth, the higher growth in 2013 indicates stronger year-over-year expansion.

3. Business Insights

- The company generated 12.64 million in sales and 1.47 million in profit, with an overall profit margin of 11.61%.
- Technology is the strongest category because it leads both sales and profit, making it a key area for continued investment and inventory planning.
- Central is the strongest region in both sales and profit, making it a useful benchmark for identifying successful regional strategies.
- Consumer customers are the largest sales-contributing segment, so targeted retention and promotional strategies for this segment may have a significant impact on revenue.
- December is the strongest sales month while February is the weakest, indicating that inventory, staffing, and marketing should be planned around seasonal demand.
- Copiers is the most profitable sub-category, while Tables produces a negative profit contribution and requires further investigation into pricing, discounts, and costs.
- Positive year-over-year growth occurred throughout the analyzed period, with the strongest growth of 27.2% occurring in 2013.

4. Actionable Recommendations

- Prioritize Technology products in inventory planning and marketing because the category leads both sales and profit.
- Study the practices behind Central's strong performance and evaluate whether successful approaches can be adapted to weaker regions.
- Develop targeted campaigns and retention initiatives for the Consumer segment because it contributes the largest share of sales.
- Prepare inventory and promotional campaigns ahead of December to capture peak demand while using targeted offers or campaigns to improve February performance.
- Protect and expand profitable sub-categories such as Copiers while reviewing the cost structure, pricing, discounts, and product mix of Tables.
- Track sales growth annually using year-over-year KPIs so that changes in performance can be identified early.
- Use profit margin alongside sales as a core management KPI to avoid optimizing for revenue without considering profitability.

5. Power BI Dashboard

An interactive Power BI dashboard was designed to communicate the analysis to business users. The dashboard includes KPI cards, trend analysis, category and regional comparisons, top-product analysis, customer segment analysis, and interactive filters.

- KPI Cards: Total Sales, Total Profit, Total Quantity Sold, Profit Margin
- Monthly Sales Trend
- Sales by Category

- Sales by Region
- Top 10 Products by Sales
- Sales by Customer Segment
- Profit by Sub-Category
- Slicers: Year, Region, Category, Segment

6. Statistical and Analytical Methods

The project used descriptive and comparative analytical methods rather than inferential statistical testing. The main methods were aggregation by business dimensions, KPI calculation, time-series grouping, year-over-year percentage growth, profit-margin calculation, ranking, and comparison of sales and profit.

- Sum aggregation for sales, profit, and quantity.
- Grouping and ranking to identify top products, categories, regions, and segments.
- Monthly and yearly trend analysis.
- Year-over-year percentage growth analysis.
- Profit margin calculation.
- Sales-versus-profit comparison to distinguish revenue performance from profitability.

Conclusion

The analysis provides a clear view of the company's sales and profitability performance. The business generated 12.64 million in sales and 1.47 million in profit, with an 11.61% overall profit margin. Technology and the Central region were the strongest contributors, while the Consumer segment generated the highest sales. December represented the peak sales period, whereas February was the weakest. Copiers was the most profitable sub-category, while Tables generated a loss. The findings can be used to improve inventory planning, regional strategy, customer targeting, seasonal campaigns, and product-level profitability management.

Project Deliverables

- Python/Jupyter Notebook containing data cleaning and exploratory analysis.
- Cleaned CSV dataset: Global_Superstore_Cleaned.csv
- Power BI dashboard: Retail Sales Performance Dashboard
- Project report documenting methodology, findings, and recommendations.